

Weyl's Criterion

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definitions

- equidistribution

We say that a sequence of real numbers (x_n) is equidistributed modulo one when for any $M \in \mathbb{Z}_{>0}$ and for any integer $0 \leq i \leq M-1$,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \# \left\{ 1 \leq j \leq N : \{x_j\} \in \left[\frac{i}{M}, \frac{i+1}{M} \right) \right\} = \frac{1}{M}.$$

In other words, $\forall \epsilon > 0$ and $M \in \mathbb{Z}_{>0}$, $\exists N_0 > 0$ s.t. $\forall N > N_0$ and $0 \leq i \leq M-1$,

$$\left| \frac{1}{N} \# \left\{ 1 \leq j \leq N : \{x_j\} \in \left[\frac{i}{M}, \frac{i+1}{M} \right) \right\} - \frac{1}{M} \right| < \epsilon.$$

- trigonometric polynomial

Trigonometric polynomial $P(x)$ of degree N could be expressed as

$$P(x) = \sum_{n=-N}^N c_n e^{inx} \text{ where } c_n \in \mathbb{C}.$$

Weyl's Criterion:

$$(x_n) \text{ is equidistributed modulo one} \iff \frac{1}{N} \sum_{n \leq N} e(hx_n) \longrightarrow 0$$

for every $h \in \mathbb{Z} \setminus \{0\}$

proof.

($\Rightarrow$) Assume (x_n) is equidistributed modulo one. By definition, $\forall \epsilon > 0$ and $M \in \mathbb{Z}_{>0}$, $\exists N_0 > 0$ s.t. $\forall N > N_0$ and $0 \leq i \leq M-1$, $|\# \{1 \leq j \leq N : \{x_j\} \in [\frac{i}{M}, \frac{i+1}{M})\} - \frac{N}{M}| < N\epsilon$. For simplicity, let's define $S_i := \{1 \leq j \leq N : \{x_j\} \in [\frac{i}{M}, \frac{i+1}{M})\}$. So we have $\forall \epsilon > 0$ and $M \in \mathbb{Z}_{>0}$, $\exists N_0 > 0$ s.t. $\forall N > N_0$ and $0 \leq i \leq M-1$, $||S_i| - \frac{N}{M}| < N\epsilon$. Observe that S_i are disjoint and cover $\{1, 2, \dots, N\}$. So,

$$\begin{aligned} \left| \frac{1}{N} \sum_{n \leq N} e(hx_n) - \frac{1}{M} \sum_{i=0}^{M-1} e\left(\frac{hi}{M}\right) \right| &= \left| \frac{1}{N} \sum_{i=0}^{M-1} \sum_{n \in S_i} e(hx_n) - \frac{1}{M} \sum_{i=0}^{M-1} e\left(\frac{hi}{M}\right) \right| \\ &\leq \sum_{i=0}^{M-1} \left| \frac{1}{N} \sum_{n \in S_i} e(hx_n) - \frac{1}{M} e\left(\frac{hi}{M}\right) \right| \\ &= \frac{1}{N} \sum_{i=0}^{M-1} \left| \sum_{n \in S_i} e(hx_n) - \frac{N}{M} e\left(\frac{hi}{M}\right) \right| \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{N} \sum_{i=0}^{M-1} \left| \sum_{n \in S_i} \left(e(hx_n) - e\left(\frac{hi}{M}\right) + e\left(\frac{hi}{M}\right) \right) - \frac{N}{M} e\left(\frac{hi}{M}\right) \right| \\
&= \frac{1}{N} \sum_{i=0}^{M-1} \left| \sum_{n \in S_i} \left(e(hx_n) - e\left(\frac{hi}{M}\right) \right) + \sum_{n \in S_i} e\left(\frac{hi}{M}\right) - \frac{N}{M} e\left(\frac{hi}{M}\right) \right| \\
&= \frac{1}{N} \sum_{i=0}^{M-1} \left| \sum_{n \in S_i} \left(e(hx_n) - e\left(\frac{hi}{M}\right) \right) + |S_i| e\left(\frac{hi}{M}\right) - \frac{N}{M} e\left(\frac{hi}{M}\right) \right| \\
&= \frac{1}{N} \sum_{i=0}^{M-1} \left| \sum_{n \in S_i} \left(e(hx_n) - e\left(\frac{hi}{M}\right) \right) + e\left(\frac{hi}{M}\right) \left(|S_i| - \frac{N}{M} \right) \right|.
\end{aligned}$$

Since $\frac{i}{M} \leq \{x_n\} < \frac{i+1}{M}$, $0 \leq \{x_n\} - \frac{i}{M} < \frac{1}{M}$. So $|\{x_n\} - \frac{i}{M}| < \frac{1}{M}$. Thus $|h\{x_n\} - \frac{hi}{M}| < \frac{|h|}{M}$. Since $|e(hx_n) - e(\frac{hi}{M})| = |e(h\{x_n\}) - e(\frac{hi}{M})| \leq 2\pi |h\{x_n\} - \frac{hi}{M}| < 2\pi \frac{|h|}{M}$, $|e(hx_n) - e(\frac{hi}{M})| < 2\pi \frac{|h|}{M}$. So,

$$\begin{aligned}
\left| \frac{1}{N} \sum_{n \leq N} e(hx_n) - \frac{1}{M} \sum_{i=0}^{M-1} e\left(\frac{hi}{M}\right) \right| &\leq \frac{1}{N} \sum_{i=0}^{M-1} \left(\sum_{n \in S_i} \left| e(hx_n) - e\left(\frac{hi}{M}\right) \right| + \left| e\left(\frac{hi}{M}\right) \right| \left| |S_i| - \frac{N}{M} \right| \right) \\
&< \frac{1}{N} \sum_{i=0}^{M-1} \left(\sum_{n \in S_i} 2\pi \frac{|h|}{M} + \left| |S_i| - \frac{N}{M} \right| \right) \\
&= \frac{1}{N} \sum_{i=0}^{M-1} \left(|S_i| 2\pi \frac{|h|}{M} + \left| |S_i| - \frac{N}{M} \right| \right) \\
&= \frac{1}{N} \sum_{i=0}^{M-1} |S_i| 2\pi \frac{|h|}{M} + \frac{1}{N} \sum_{i=0}^{M-1} \left| |S_i| - \frac{N}{M} \right|.
\end{aligned}$$

Since $\forall \epsilon > 0$ and $M \in \mathbb{Z}_{>0}$, $\exists N_0 > 0$ s.t. $\forall N > N_0$ and $0 \leq i \leq M-1$, $\left| |S_i| - \frac{N}{M} \right| < N\epsilon$,

$$\begin{aligned}
\left| \frac{1}{N} \sum_{n \leq N} e(hx_n) - \frac{1}{M} \sum_{i=0}^{M-1} e\left(\frac{hi}{M}\right) \right| &< \frac{1}{N} \sum_{i=0}^{M-1} |S_i| 2\pi \frac{|h|}{M} + \frac{1}{N} \sum_{i=0}^{M-1} (N\epsilon) \\
&= \frac{1}{N} \sum_{i=0}^{M-1} |S_i| 2\pi \frac{|h|}{M} + M\epsilon \\
&< \frac{1}{N} \sum_{i=0}^{M-1} \left(\frac{N}{M} + N\epsilon \right) 2\pi \frac{|h|}{M} + M\epsilon \\
&= \sum_{i=0}^{M-1} \left(\left(\frac{1}{M} + \epsilon \right) 2\pi \frac{|h|}{M} \right) + M\epsilon \\
&= M \left(\frac{1}{M} + \epsilon \right) 2\pi \frac{|h|}{M} + M\epsilon \\
&= \left(\frac{1}{M} + \epsilon \right) 2\pi |h| + M\epsilon.
\end{aligned}$$

Observe

$$\sum_{i=0}^{M-1} e\left(\frac{hi}{M}\right) = \sum_{i=0}^{M-1} e^{2\pi i \frac{hi}{M}} = \sum_{i=0}^{M-1} \left(e^{2\pi i \frac{h}{M}} \right)^i = \frac{1 - e^{2\pi i \frac{h}{M} M}}{1 - e^{2\pi i \frac{h}{M}}} = \frac{1 - e^{2\pi i h}}{1 - e^{2\pi i \frac{h}{M}}}.$$

Since $h \in \mathbb{Z} \setminus \{0\}$, $\sum_{i=0}^{M-1} e(\frac{hi}{M}) = 0$ when $M \nmid h$. Later, we will consider large M so it is enough to know that

$$\sum_{i=0}^{M-1} e(\frac{hi}{M}) = 0 \text{ for } M > |h|.$$

So, the inequality simplifies to

$$\left| \frac{1}{N} \sum_{n \leq N} e(hx_n) \right| < \frac{2\pi|h|}{M} + \epsilon(2\pi|h| + M). \quad (1)$$

Let's fix some $\delta > 0$. Now fix an integer $M > |h|$ large enough so that $\frac{2\pi|h|}{M} < \frac{\delta}{2}$. Then fix ϵ small enough so that $\epsilon(2\pi|h| + M) < \frac{\delta}{2}$. We know that $\forall \epsilon > 0$ and $M \in \mathbb{Z}_{>0}$, $\exists N_0 > 0$ s.t. $\forall N > N_0$ the bound (1) holds. Therefore, for our chosen ϵ and M , $\exists N_0 > 0$ s.t. $\forall N > N_0$,

$$\left| \frac{1}{N} \sum_{n \leq N} e(hx_n) \right| < \frac{\delta}{2} + \frac{\delta}{2} = \delta.$$

So we can conclude

$$\frac{1}{N} \sum_{n \leq N} e(hx_n) \longrightarrow 0.$$

($\Leftarrow$) Assume $\frac{1}{N} \sum_{n \leq N} e(hx_n) \longrightarrow 0$. The trigonometric Weierstrass approximation theorem states that for any continuous periodic function $f : \mathbb{R} \rightarrow \mathbb{C}$ of period 2π , and for any $\epsilon > 0$, there exists a trigonometric polynomial P such that $\sup_{x \in \mathbb{R}} |f(x) - P(x)| < \epsilon$. So for any continuous periodic function $f : \mathbb{R} \rightarrow \mathbb{C}$ of period 1, and for any $\epsilon > 0$, there exists a trigonometric polynomial P such that $\sup_{x \in \mathbb{R}} |f(x) - P(x)| < \epsilon$ where here, P can be expressed as $\mathbb{C}$ -linear combination of finitely many $e(hx)$. So,

$$\begin{aligned} \left| \frac{1}{N} \sum_{n=1}^N f(x_n) - \int_0^1 f(x) dx \right| &\leq \left| \frac{1}{N} \sum_{n=1}^N (f(x_n) - P(x_n)) \right| + \left| \frac{1}{N} \sum_{n=1}^N P(x_n) - \int_0^1 P(x) dx \right| + \left| \int_0^1 (P(x) - f(x)) dx \right| \\ &< \epsilon + \left| \frac{1}{N} \sum_{n=1}^N P(x_n) - \int_0^1 P(x) dx \right| + \epsilon \\ &= 2\epsilon + \left| \frac{1}{N} \sum_{n=1}^N P(x_n) - \int_0^1 P(x) dx \right|. \end{aligned}$$

Let $c_a \in \mathbb{C}$. Since $\frac{1}{N} \sum_{n \leq N} e(hx_n) \longrightarrow 0$ for $h \in \mathbb{Z} \setminus \{0\}$,

$$\frac{1}{N} \sum_{n=1}^N P(x_n) = \frac{1}{N} \sum_{n=1}^N \sum_{-k \leq a \leq k} c_a e(ax_n) \xrightarrow{N \rightarrow \infty} c_0,$$

$$\begin{aligned} \int_0^1 P(x) dx &= \int_0^1 \sum_{-k \leq a \leq k} c_a e(ax) dx = \sum_{-k \leq a \leq k} \int_0^1 c_a e(ax) dx = \int_0^1 c_0 e(0) dx + 0 \\ &= c_0 x \Big|_0^1 \\ &= c_0. \end{aligned}$$

Therefore,

$$\left| \frac{1}{N} \sum_{n=1}^N P(x_n) - \int_0^1 P(x) dx \right| \xrightarrow{N \rightarrow \infty} 0.$$

Hence, given our initial assumption, for any 1-periodic continuous function f ,

$$\frac{1}{N} \sum_{n=1}^N f(x_n) \longrightarrow \int_0^1 f(x) dx \text{ as } N \rightarrow \infty.$$

lemma If for every 1-periodic continuous function $f : \mathbb{R} \rightarrow \mathbb{C}$ we have

$$\frac{1}{N} \sum_{n=1}^N f(x_n) \longrightarrow \int_0^1 f(x) dx \text{ as } N \rightarrow \infty,$$

then the sequence of real numbers (x_n) is equidistributed modulo one.

Proof. Let $[a, b] \subseteq [0, 1)$ be an arbitrary interval and fix $\epsilon > 0$. We can choose 1-periodic continuous functions f_- and $f_+ : \mathbb{R} \rightarrow \mathbb{R}_{\geq 0}$ such that

$$0 \leq f_- \leq \mathbf{1}_{[a,b]} \leq f_+$$

where the area under each curve is within ϵ of the interval's length, $b - a$:

$$\int_0^1 f_-(x) dx > b - a - \epsilon \quad \text{and} \quad \int_0^1 f_+(x) dx < b - a + \epsilon.$$

Observe that

$$\begin{aligned} \frac{1}{N} \#\{1 \leq n \leq N : x_n \in [a, b]\} &= \frac{1}{N} \sum_{n=1}^N \mathbf{1}_{[a,b]}(x_n) \\ &\leq \frac{1}{N} \sum_{n=1}^N f_+(x_n) \xrightarrow{N \rightarrow \infty} \int_0^1 f_+(x) dx < b - a + \epsilon. \end{aligned}$$

So,

$$\limsup_{N \rightarrow \infty} \frac{1}{N} \#\{1 \leq n \leq N : x_n \in [a, b]\} \leq b - a + \epsilon.$$

Similarly,

$$\begin{aligned} \frac{1}{N} \#\{1 \leq n \leq N : x_n \in [a, b]\} &= \frac{1}{N} \sum_{n=1}^N \mathbf{1}_{[a,b]}(x_n) \\ &\geq \frac{1}{N} \sum_{n=1}^N f_-(x_n) \xrightarrow{N \rightarrow \infty} \int_0^1 f_-(x) dx > b - a - \epsilon. \end{aligned}$$

So,

$$\liminf_{N \rightarrow \infty} \frac{1}{N} \#\{1 \leq n \leq N : x_n \in [a, b]\} \geq b - a - \epsilon.$$

Therefore,

$$\lim_{N \rightarrow \infty} \frac{1}{N} \#\{1 \leq n \leq N : x_n \in [a, b]\} = b - a.$$

Since this holds for any arbitrary interval $[a, b] \subseteq [0, 1)$, we can conclude by definition that the sequence (x_n) is equidistributed modulo 1. $\square$